

People's Democratic Republic of Algeria
Ministry of Higher Education
and Scientific Research
Higher National School of Biotechnology "Toufik Khaznadar"



Machine Learning Application in Drug Discovery : QSAR

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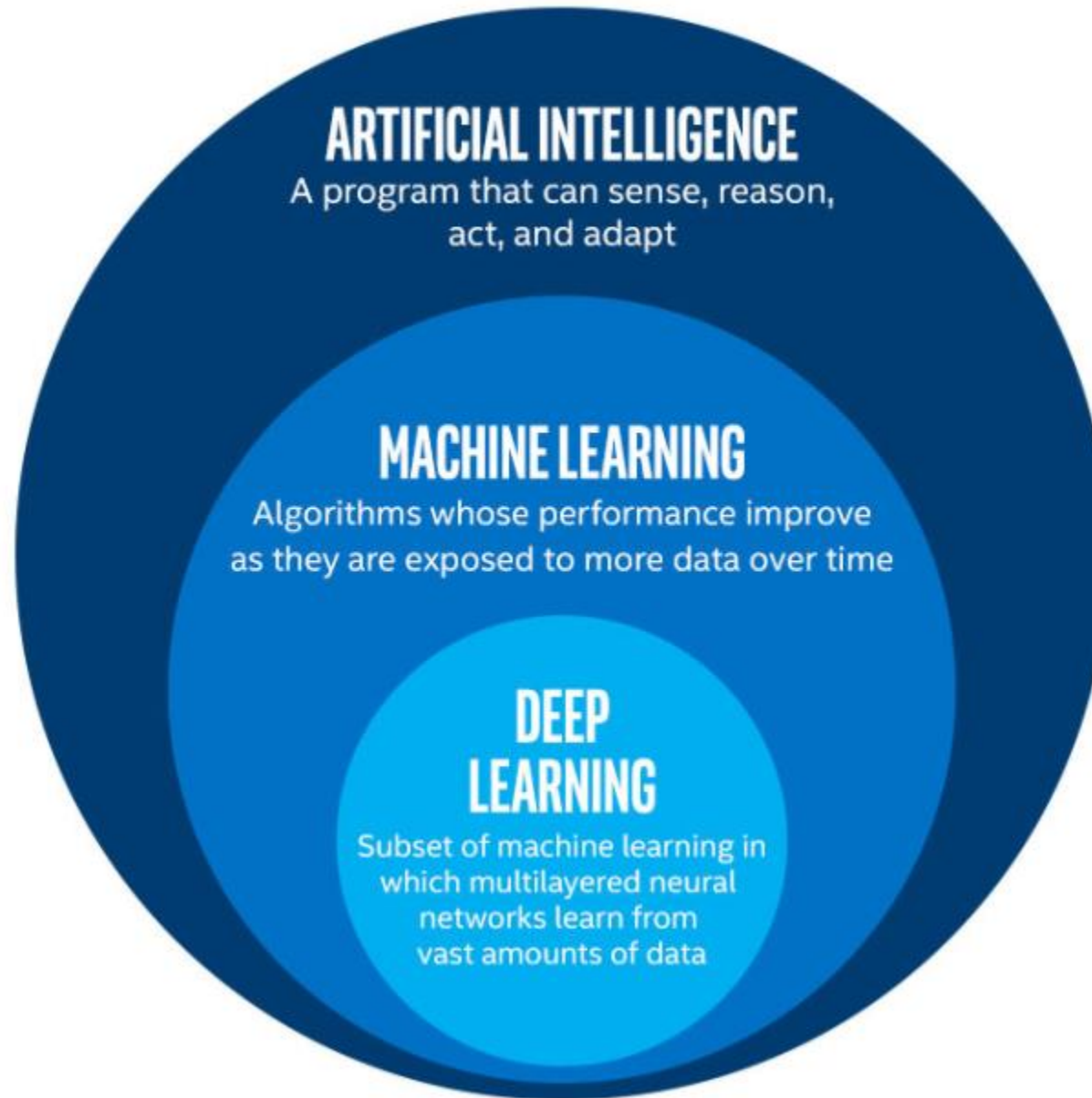
Exploratory Data Analysis under Lipinski's Rule

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Molecular Descriptors

What is Machine Learning ?

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What is Machine Learning ?

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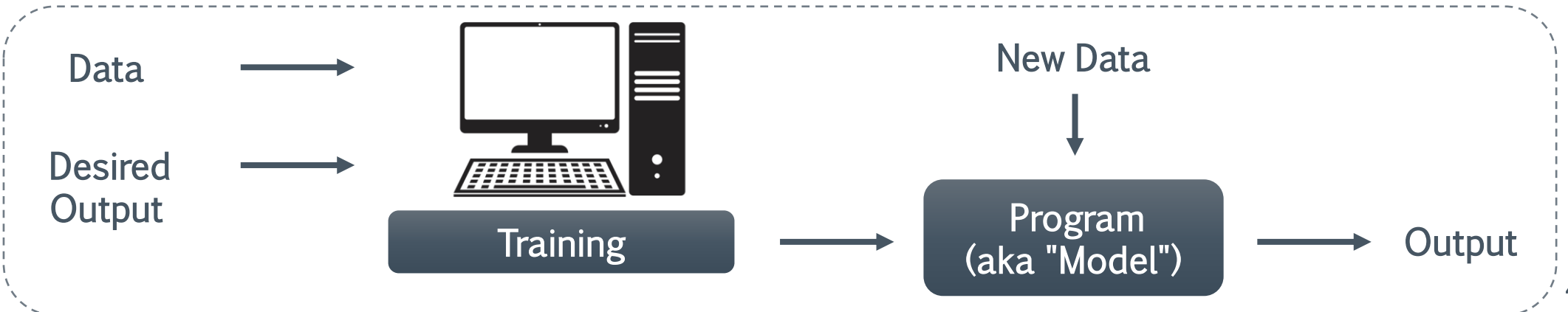
Traditional Programming

- Developers write rules (program) that produce an output.



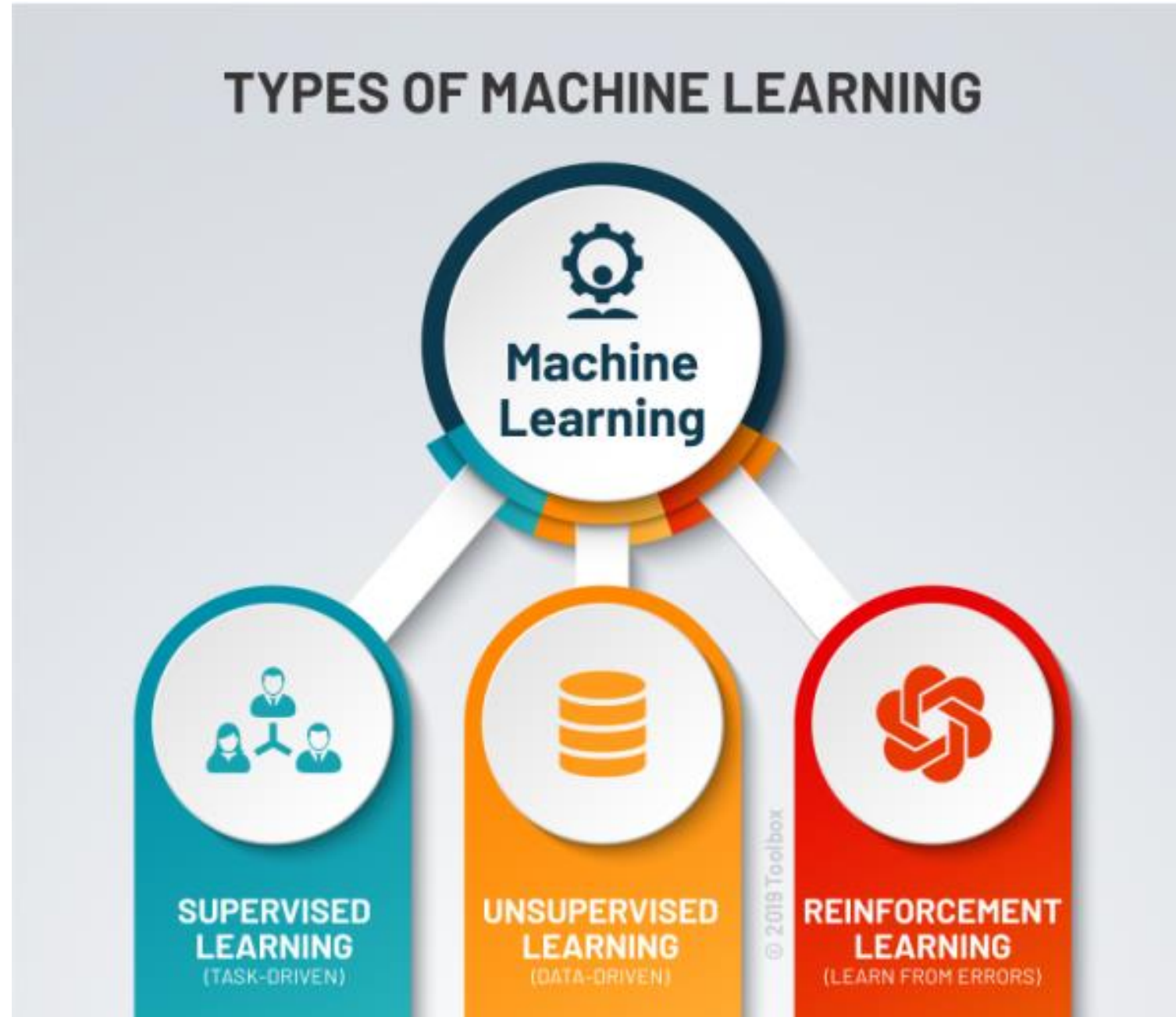
Machine Learning

- Developers write a training algorithm, that finds rules, which produce the desired output.



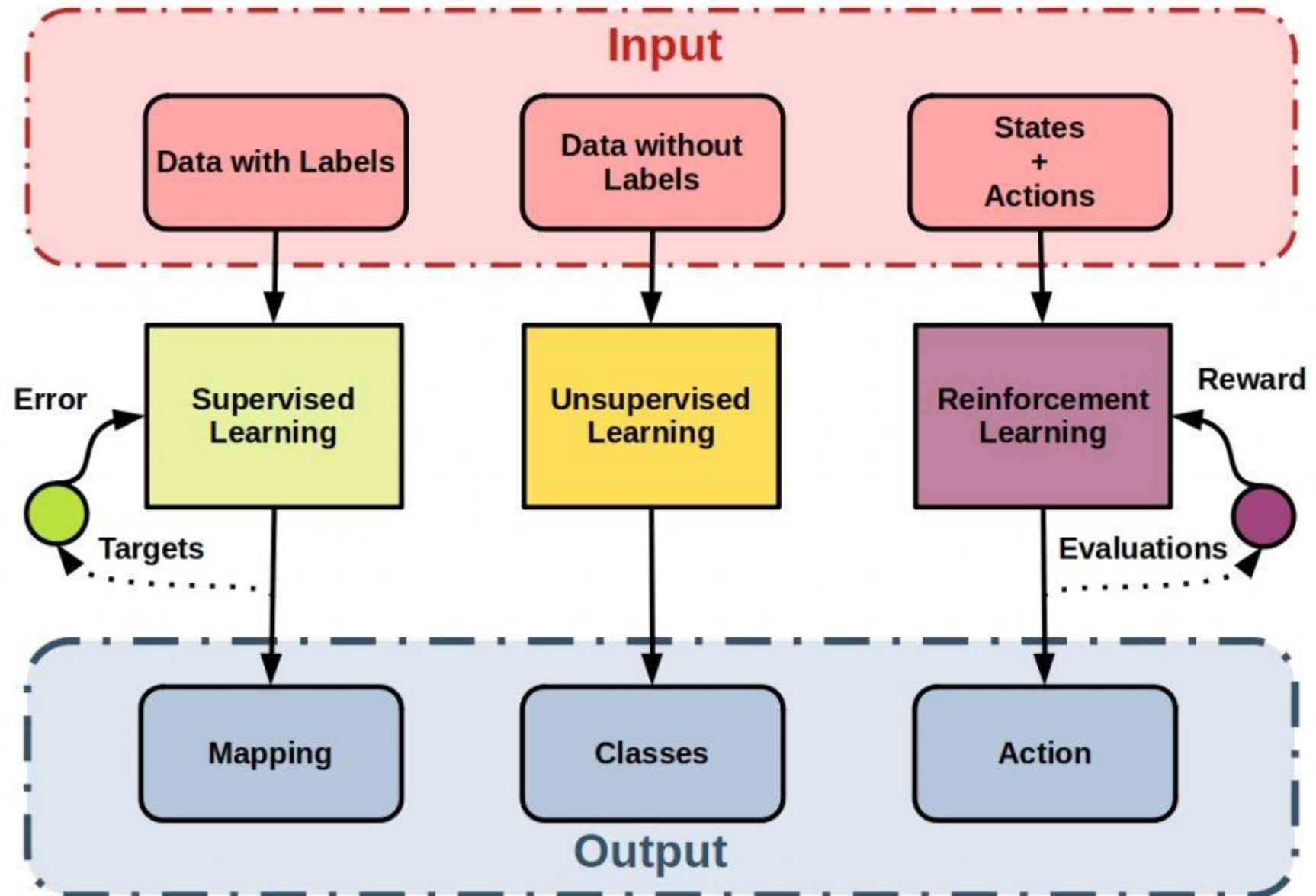
Machine Learning Types

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Machine Learning Types

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Supervised vs Unsupervised Learning

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Supervised

X_1	X_2	...	X_n	Y

Independent
Variables

Target
Dependent
Variable

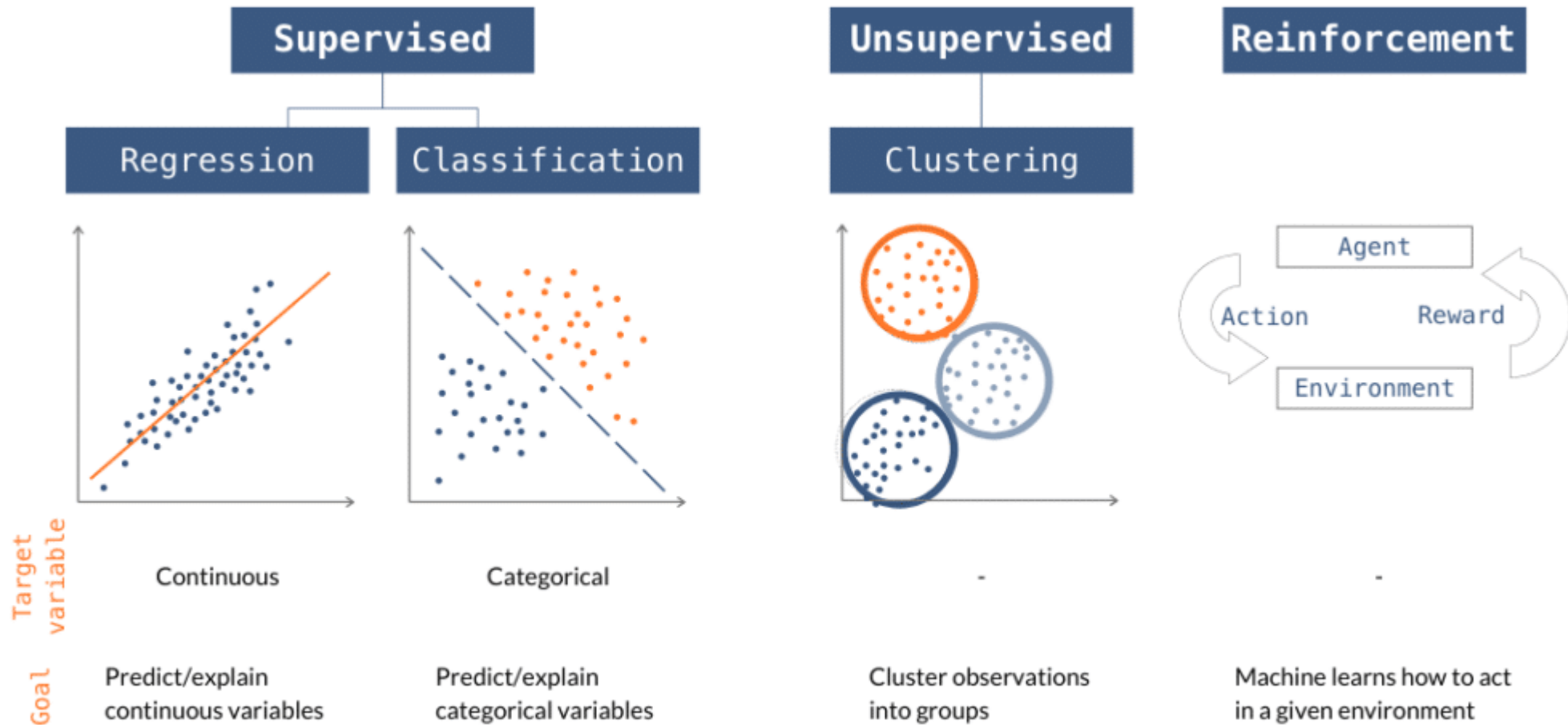
Un-Supervised

X_1	X_2	...	X_n	Y

No
Target

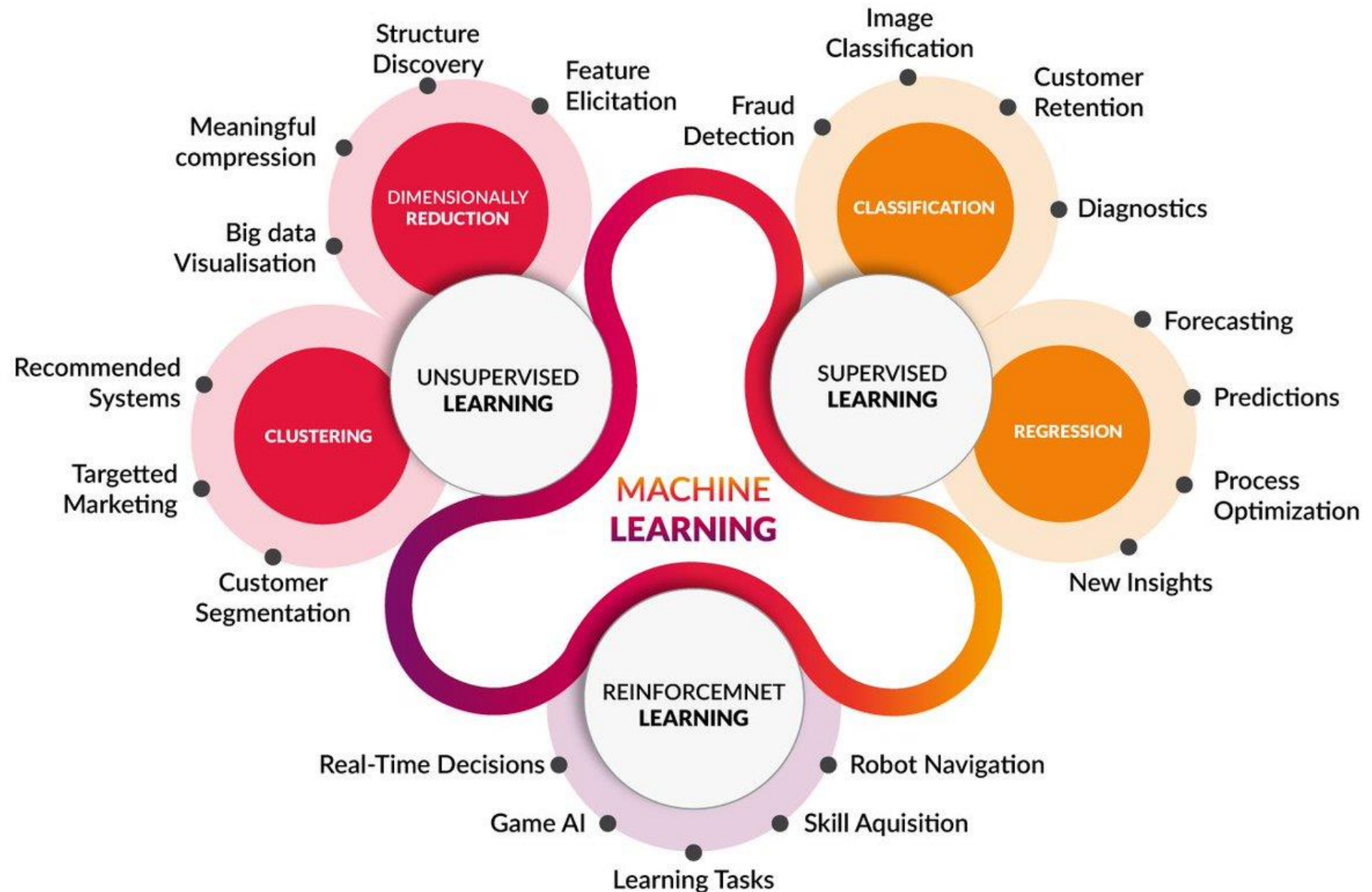
Supervised vs Unsupervised vs Reinforcement

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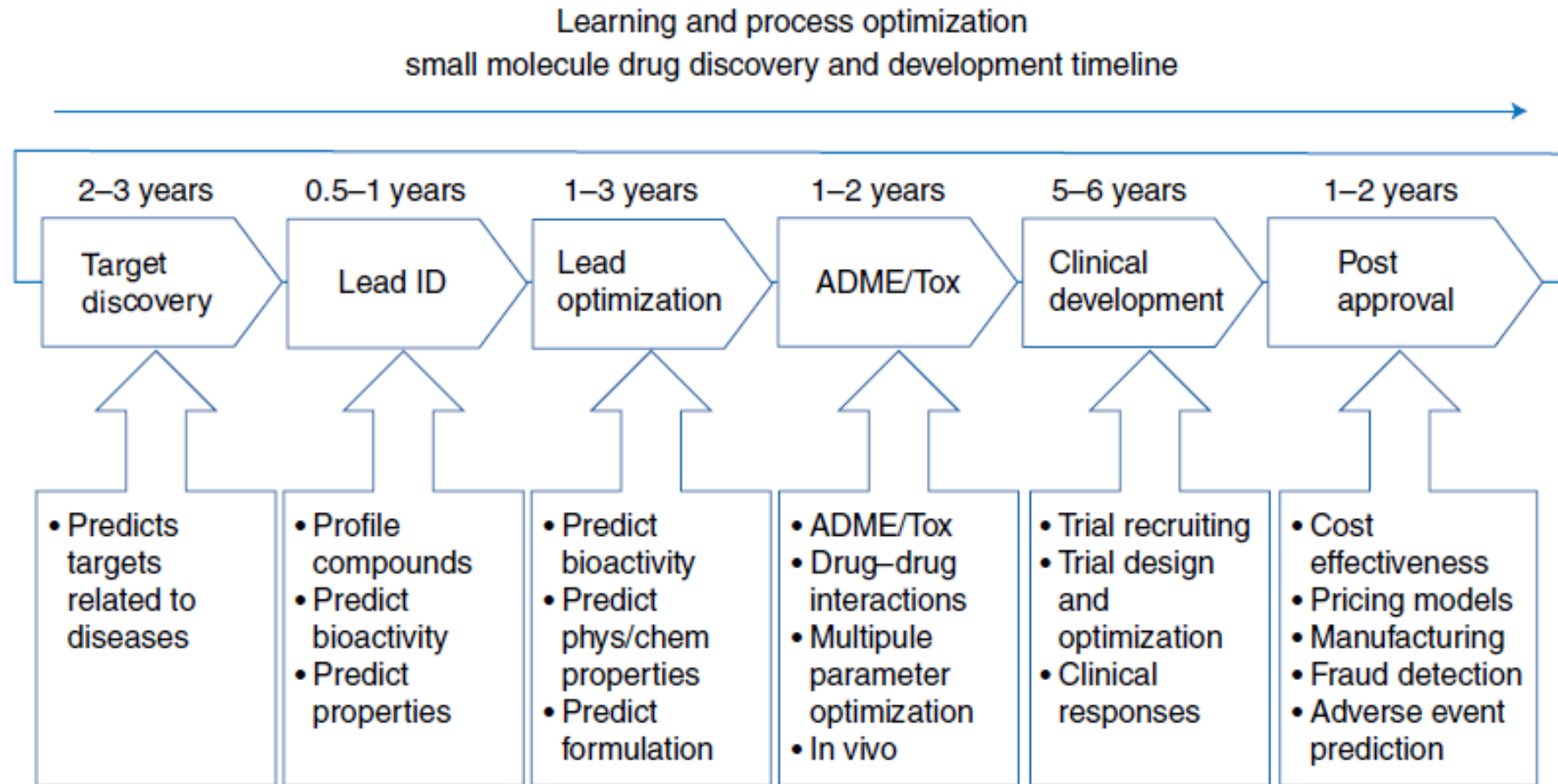
Machine Learning Applications

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Stages of Drug Discovery and Development

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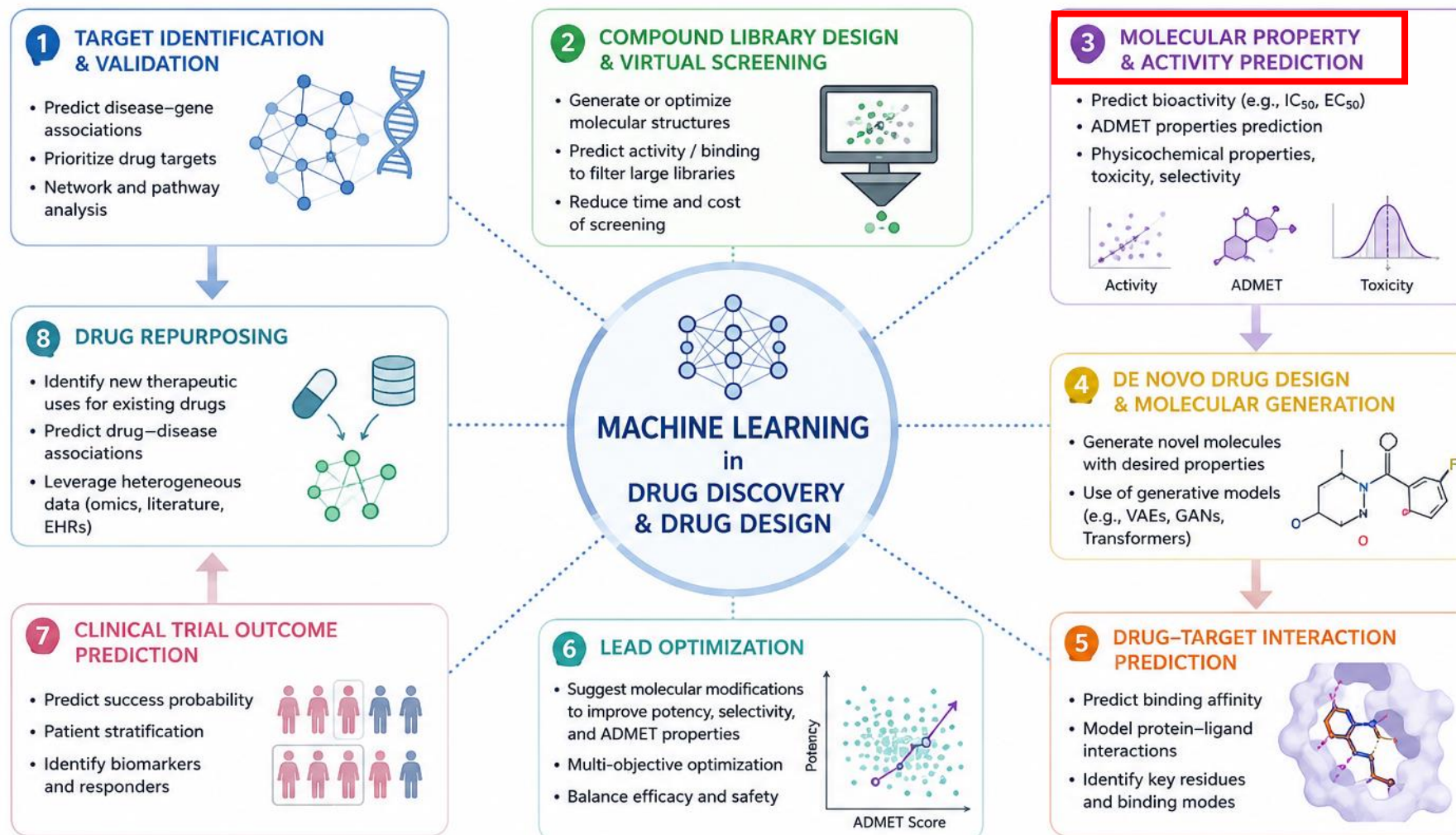


Reference : Ekins, S., Puhl, A.C., Zorn, K.M. et al. Exploiting machine learning for end-to-end drug discovery and development. Nat. Mater. 18, 435–441 (2019).

Machine Learning Applications in Drug Discovery

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Applications of Machine Learning in Drug Discovery / Drug Design



BENEFITS



Faster discovery and development



Lower costs



Improved success rates



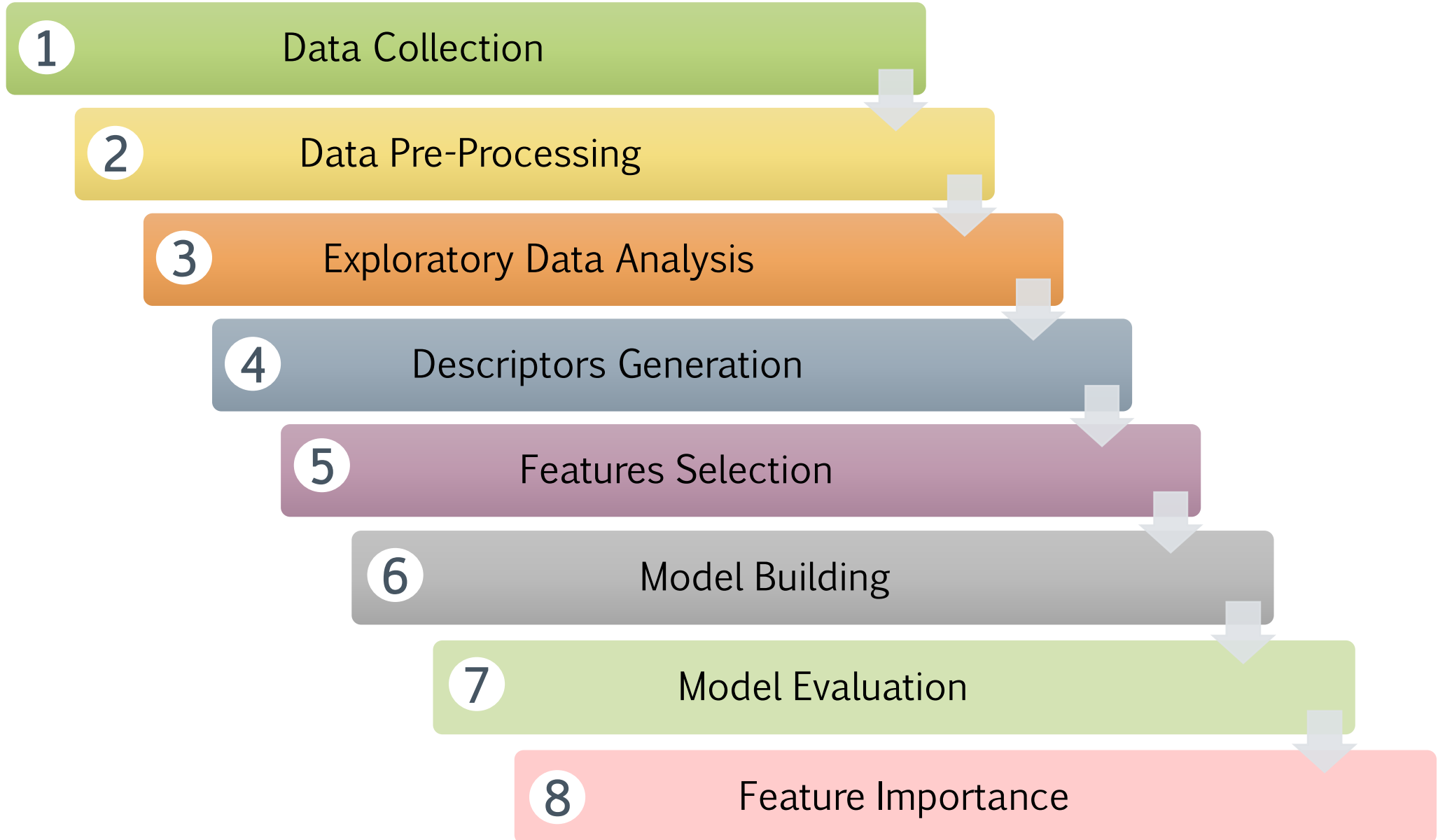
Better safety assessment



More personalized medicines

QSAR Application : Machine Learning Pipeline

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Python Libraries for Machine Learning

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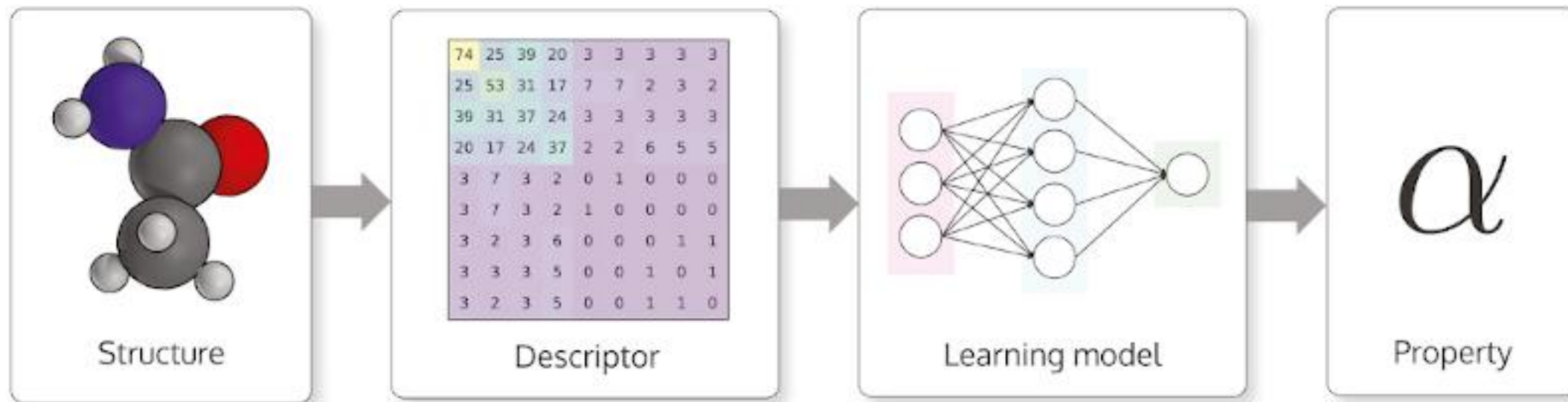


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Molecule Representation

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SMILES : Simplified Molecular Input Line Entry Specification

Before doing Exploratory Data Analysis

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Why convert IC50 to pIC50?

pIC50 is the standard scale in QSAR to measure biological activity

1 Raw IC50 values: very wide range

IC50 values often span several orders of magnitude (logarithmic scale).

Molecule	IC50 (nM)
A	1
B	10
C	100
D	1000

$\times 10$
 $\times 10$
 $\times 10$



Difficult for statistical analysis and machine learning

2 Logarithmic transformation to pIC50

$$pIC50 = -\log_{10}(\text{IC50 in molar})$$

IC50 (nM)	pIC50
1	9
10	8
100	7
1000	6

-1
 -1
 -1



The logarithmic transformation compresses the scale and reduces data dispersion.

3 Advantages of pIC50



Reduces data dispersion and the influence of extreme values



More suitable for ML models improves performance of machine learning algorithms



More intuitive interpretation and biologically consistent

Low IC50 → highly active molecule (e.g., 1 nM)

High pIC50 → highly active molecule (e.g., 9)

Illustration: increasing activity

IC50 (nM)	100,000 nM (100 μ M)	1000 nM (1 μ M)	100 nM	10 nM	1 nM
pIC50	4	6	7	8	9



Key takeaway

In QSAR, pIC50 is generally preferred over raw IC50 because it provides a more stable logarithmic scale, reduces the impact of outliers, is easier to interpret, and is better suited for machine learning.



Christopher Lipinski, a scientist at Pfizer, came up with a set of rule-of-thumb for evaluating the **druglikeness** of compounds. Such druglikeness is based on the Absorption, Distribution, Metabolism and Excretion (ADME) that is also known as the pharmacokinetic profile. Lipinski analyzed all orally active FDA-approved drugs in the formulation of what is to be known as the **Rule-of-Five** or **Lipinski's Rule**.

The Lipinski's Rule stated the following:

- Molecular weight < 500 Dalton
- Octanol-water partition coefficient (LogP) < 5
- Hydrogen bond donors < 5
- Hydrogen bond acceptors < 10

Exploratory Data Analysis under Lipinski's Rule

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LIPINSKI'S RULE OF FIVE

A practical rule to evaluate oral drug-likeness



Goal: predict whether a molecule has good chances of being well absorbed when taken orally.

Lipinski's rules estimate if a compound has physicochemical properties compatible with good oral bioavailability.

- ✓ Absorbed by the gastrointestinal tract
- ✓ Able to cross biological membranes
- ✓ Compatible with standard pharmaceutical development



Not a rule of biological activity (does not predict if it works).



A rule of bioavailability (does it reach the target?).



KEY IDEA

A good oral drug must find the right balance between: **water solubility** (in blood) and **membrane permeability** (lipids).

THE 4 CRITERIA AND THEIR PURPOSE

1

MOLECULAR WEIGHT
 ≤ 500 Da



Controls the size

- ✓ Too large = difficult to absorb
- ✓ Smaller = better diffusion

2

LogP
 ≤ 5

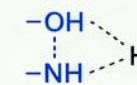


Controls lipophilicity

- ✓ Balance between water and lipids
- ✗ Too lipophilic = poorly soluble in blood
- ✗ Too hydrophilic = does not cross membranes

3

HYDROGEN BOND DONORS (HBD)
 ≤ 5

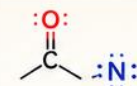


Controls interactions with water

- ✓ Too many HBD = molecule "sticks" to water
- ✓ Fewer HBD = better cell penetration

4

HYDROGEN BOND ACCEPTORS (HBA)
 ≤ 10



Controls overall polarity

- ✓ Too many HBA = strong hydration
- ✗ Reduces membrane diffusion

1

MOLECULAR WEIGHT
 ≤ 500 Da



Controls the size
Smaller molecules = better diffusion across membranes

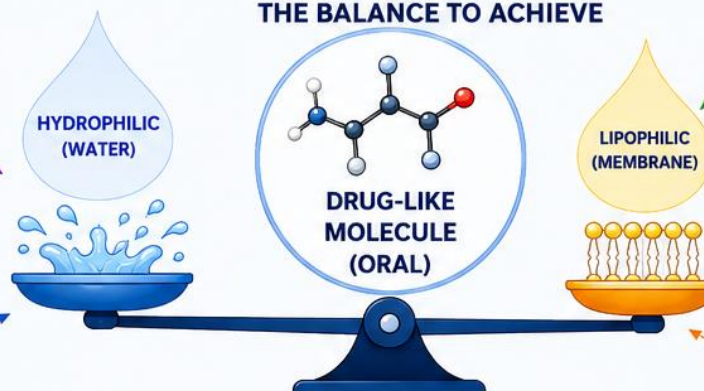
2

LogP ≤ 5



Controls lipophilicity
Balance between solubility in water and lipids (membranes)

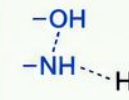
THE BALANCE TO ACHIEVE



Not too hydrophilic, not too lipophilic:
an **optimal balance** for good oral absorption!

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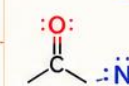
HYDROGEN BOND DONORS (HBD)
 ≤ 5



Controls interactions with water
Too many HBD = strong interaction with water = poor cell penetration

4

HYDROGEN BOND ACCEPTORS (HBA)
 ≤ 10



Controls overall polarity
Too many HBA = molecule stays more in the aqueous medium



IMPORTANT

These are not absolute rules!
Many effective drugs violate them.



USEFUL IN EDA / QSAR

- Reduce chemical space
- Identify trends
- Improve the quality of models



TAKE-HOME MESSAGE



Small enough
(≤ 500 Da)



Balanced water / lipids
(LogP ≤ 5)



Moderate interactions
with water (HBD ≤ 5 , HBA ≤ 10)

→ **Better chance of being well absorbed orally!**

Molecular Descriptors

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